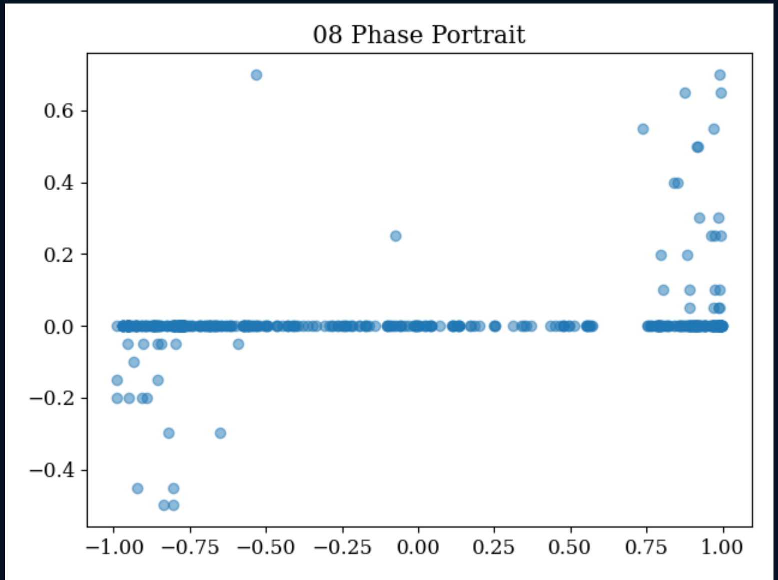
Dynamical Systems in Crypto Microstructure

A clear ODE-based explanation of high-frequency limit order books

Project by

RAJ LOHAR

Roll No: BSD-CC-2414
Centre: Indian Statistical Institute (ISI)
Email: bsdcc2414@isical.ac.in
Single-person project



Phase-space view of OBI vs price velocity

Core idea: treat the limit order book as a moving fluid system, then use ODE tools to understand liquidity, stability, shocks, memory, and execution risk.

What the project is really about

The order book is treated as a moving continuous system, not just a static table.

PROJECT IDEA

- A crypto limit order book changes extremely fast: orders are added, cancelled, matched, and shifted at sub-second scale.
- Instead of only using discrete statistical summaries, the project uses Ordinary Differential Equations to describe continuous market motion.
- The main objects are liquidity volume, bid/ask imbalance, price velocity, spread, and depth across 10 bid + 10 ask levels.
- The goal is not only prediction; it is also understanding why liquidity recovers, why shocks persist, and when price becomes unstable.

3,730,870
ticks analyzed

10 + 10
depth levels

1.1 GB
raw scale

Bitcoin
asset

Simple translation:
“This project tries to explain high-frequency crypto market behaviour using classical ODE mathematics.”

Limit order book basics

The book has two sides: bid demand and ask supply.

- Bid side: buy limit orders. Higher bid price means stronger buyer willingness.
- Ask side: sell limit orders. Lower ask price means stronger seller willingness.
- Best bid and best ask form the immediate trading boundary; their midpoint is the mid-price.
- Depth levels beyond the best quote show hidden pressure: they often move before the top price visibly changes.

$$\text{MidPrice} = (\text{Best Bid} + \text{Best Ask}) \ / \ 2$$

$$\text{Spread} = \text{Best Ask} - \text{Best Bid}$$

$$\text{OBI} = (\text{Bid Volume} - \text{Ask Volume}) \ / \ (\text{Bid Volume} + \text{Ask Volume})$$

OBI near +1 means buyers dominate.
OBI near −1 means sellers dominate.
OBI near 0 means the book is balanced.

What data was converted into ODE variables?

The project turns snapshots into state variables and derivatives.

- At every timestamp, the data contains bid prices, bid volumes, ask prices, and ask volumes across multiple levels.
- The state of the order book is treated as x(t), a vector that changes continuously with time.
- Central differences are used to approximate derivatives such as P′(t), S′(t), and liquidity velocity.
- This allows the project to ask: how fast is liquidity changing, how fast is price moving, and how do book levels influence each other?

$$\text{state at time t:}\ \ \mathrm{x(t)} = [\text{bid volumes, ask volumes, spread, velocity, imbalance, ...}]$$

$$\text{derivative} \approx (\text{value at t+}\Delta\text{t} - \text{value at t}) \ / \ \Delta\text{t}$$

Important idea: the data is still discrete, but because ticks are extremely dense, the report approximates the LOB as a continuous flow.

Overall ODE workflow used in the project

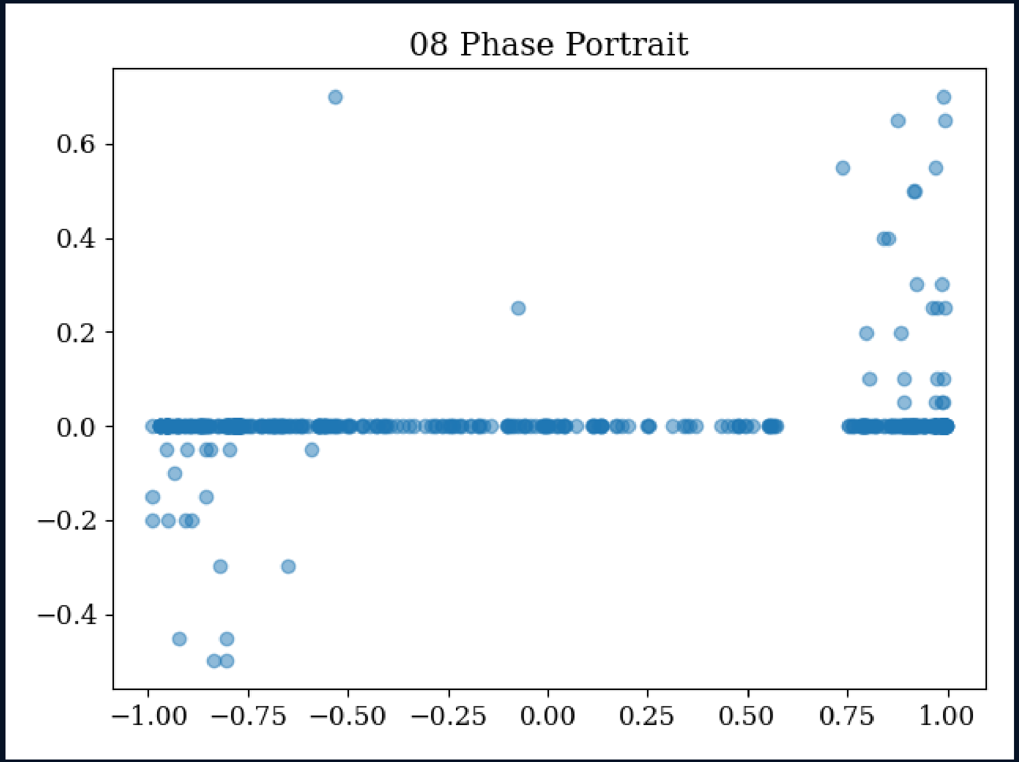
Every mathematical tool answers a different market question.

- Phase portrait → Is the book stable or does it break into directional movement?
- Logistic growth → How fast does liquidity come back after a shock?
- Wronskian → Are price and spread independent signals?
- Linear ODE systems → How do 20 depth levels move together?
- Picard iteration → Can a short price path be approximated step by step?
- Cauchy–Euler → Does shock memory decay slowly as a power law?
- Exact equations, symmetry, Lipschitz, oscillator models → slippage, bid/ask duality, uniqueness, and oscillation.

Phase portraits: the market as a moving point

A phase plot shows where the system is now and where it tends to go.

- The two main axes here are x(t) = OBI and y(t) = mid-price velocity.
- A point near (0,0) means the book is balanced and price is not moving strongly.
- A point near +1 on the OBI axis means strong buy-side pressure; a point near −1 means sell-side pressure.
- The project uses this view to study equilibrium and potential breakout behaviour.



Phase portrait of normalized OBI vs price velocity

Graph reading: most points sit near velocity = 0, but extreme OBI values create large upward/downward velocity spikes.

The coupled ODE model for OBI and price velocity

The model says imbalance creates price motion, while friction pulls both back.

$$d(\mathrm{OBI})/dt = -\,\beta \cdot \mathrm{Velocity\_y} - \gamma \cdot \mathrm{OBI}$$

$$d(\mathrm{Velocity\_y})/dt = \alpha \cdot \mathrm{OBI} - \delta \cdot \mathrm{Velocity\_y}$$

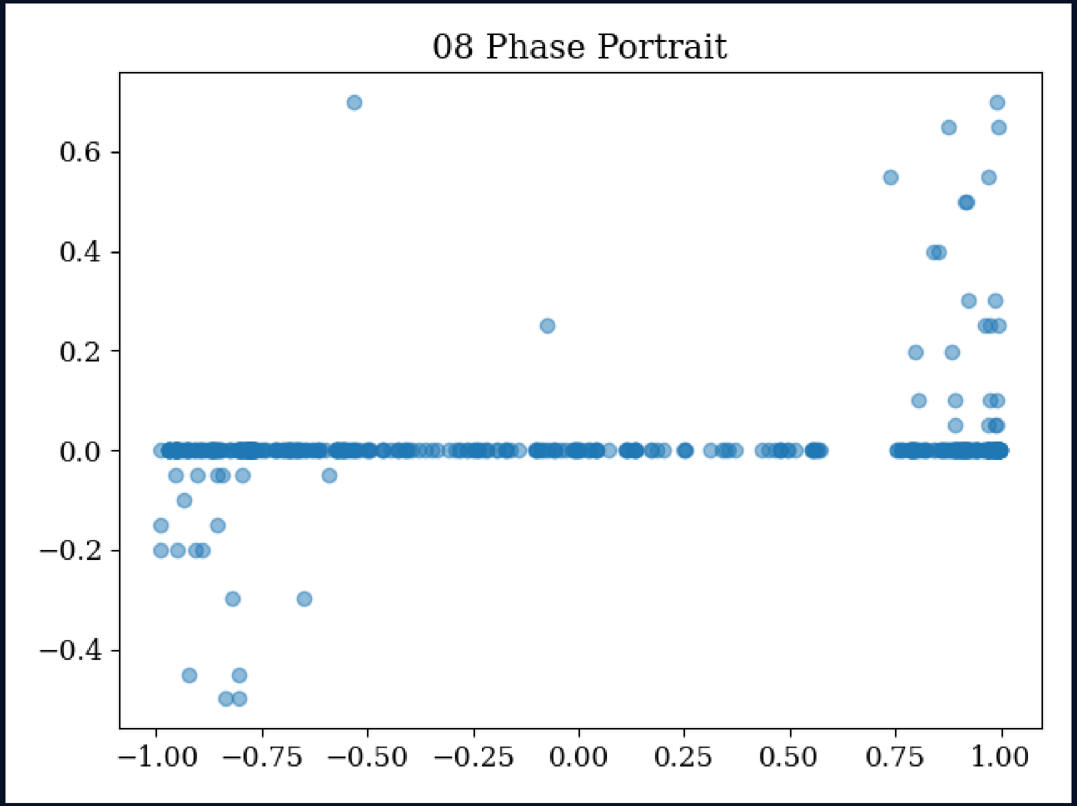
- α: how strongly imbalance pushes price velocity.
- β: how strongly price movement feeds back into imbalance.
- γ: how quickly imbalance naturally returns toward zero.
- δ: how quickly price velocity slows down due to liquidity/friction.

- Equilibrium is (OBI, Velocity) = (0,0).
- Negative eigenvalues mean return toward equilibrium.
- Positive eigenvalues mean movement away from equilibrium.
- The report finds both signs, so it interprets the microstructure as a saddle-like system.

$$A = [\, -\gamma \quad -\beta \;;\; \alpha \quad -\delta \,]$$

What the phase graph tells us

Most moments are calm, but extreme imbalance creates risk of directional motion.



OBI on x-axis, price velocity on y-axis

- The dense horizontal band at $y \approx 0$ means price usually stays stable even when OBI changes.
- At OBI near +1, upward velocity spikes appear; buy pressure can push price up.
- At OBI near −1, downward velocity spikes appear; sell pressure can pull price down.
- So the main presentation message is: extreme imbalance is more informative than small imbalance.

Careful wording: this image is a phase-space scatter plot. The saddle-point claim comes from the fitted Jacobian/eigenvalues, not from the scatter alone.

Logistic growth: liquidity returns with a limit

The order book refills after a shock, but not infinitely.

$$dV/dt = r\ V(t) \cdot (1 - V(t)/K)$$

- $V(t)$: liquidity/volume available in the book at time t.
- r: recovery rate — how fast liquidity comes back.
- K: carrying capacity — the maximum sustainable liquidity level.
- The factor (1 − V/K) slows the growth as liquidity gets close to K.

Financial intuition: after a large trade or cancellation wave, market makers refill liquidity. They act fast, but they have capital limits and risk limits, so recovery must saturate.

$$\begin{gathered}\text{small V} \ \rightarrow \text{fast growth} \\ \text{V close to K} \rightarrow \text{growth} \approx 0\end{gathered}$$

This creates the famous S-shaped recovery curve.

Where the logistic solution comes from

Start from growth, add a maximum limit, then integrate.

1) Unlimited growth: $dV/dt = rV$

2) Add remaining capacity: $dV/dt = rV(1 - V/K)$

3) Integrate: $\int dV/[V(1 - V/K)] = \int r\, dt$

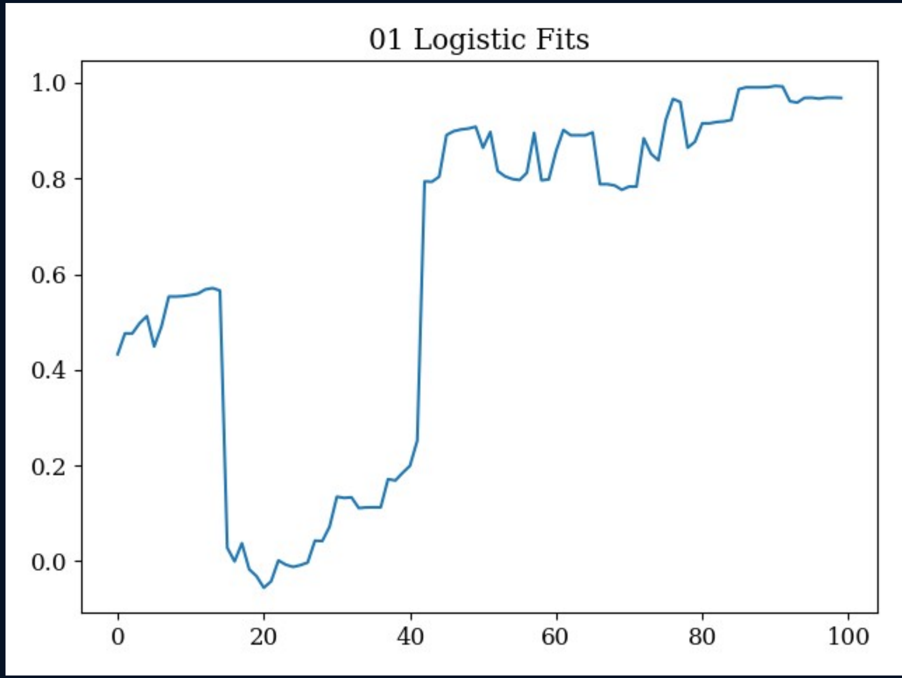
4) Final S-curve:

$$V(t) = K \;/\; [1 + ((K - V_0)/V_0)e\hat{\ }(-rt)]$$

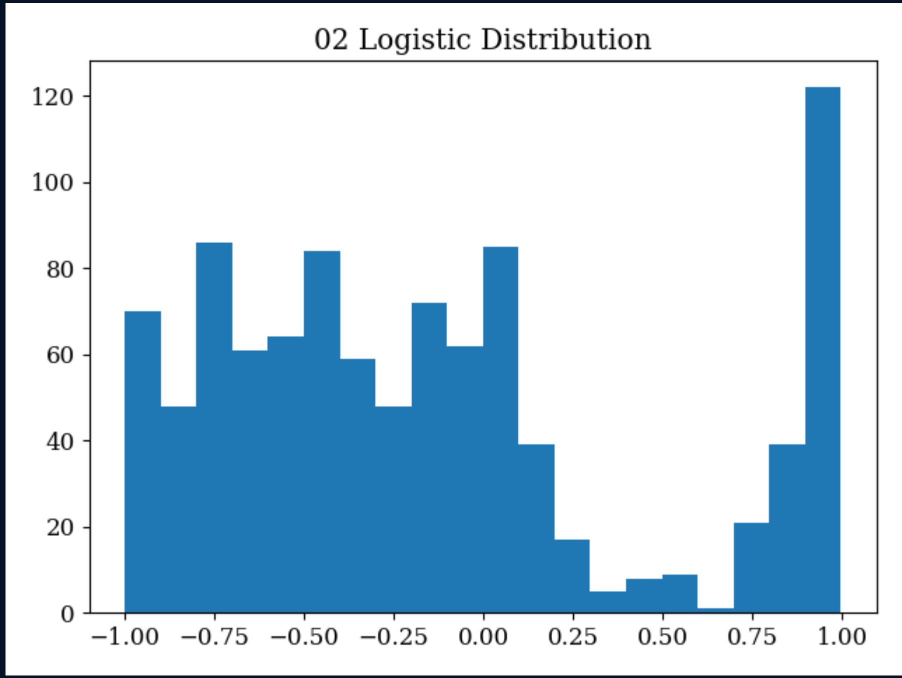
- V_0 is the initial liquidity just after the shock.
- As t increases, e^(−rt) shrinks.
- So the denominator moves toward 1 and V(t) approaches K.
- That is exactly “liquidity recovery until capacity.”

Logistic fit graphs: shock, refill, saturation

The fitted recovery speed r tells us how fast liquidity returns.



Example recovery after liquidity shock



Distribution of fitted recovery behaviour

- Left graph: liquidity drops sharply, then recovers toward a high stable level.
- The project fits 64 localized shock events using nonlinear least squares.
- Average fitted recovery rate: r = 18.32, meaning extremely fast replenishment.
- Trading interpretation: wait until liquidity crosses the logistic inflection point K/2 before aggressive execution.

Wronskian: checking if two signals are truly different

A model should not use duplicate features.

- In trading models, price, spread, volume, imbalance, and depth are used as features.
- If two features are basically the same signal, the model becomes redundant and unstable.
- The Wronskian is a calculus-based test for functional independence.
- The project checks whether price $P(t)$ and spread $S(t)$ carry independent information.

$$W(f_1,\dots,f_n)(t) = \det[\ f_i\hat{}(j-1)(t)\ ]$$

Rule used here:
If $W(t_0) \neq 0$ at at least one point, the functions are linearly independent.

Simple meaning: “they are not just copies of each other.”

Wronskian formula used for price and spread

For two signals, the determinant has a simple form.

$$W(t) = |\ P(t) \quad S(t)\ ;\ P'(t) \quad S'(t)\ |$$

$$W(t) = P(t)S'(t) - S(t)P'(t)$$

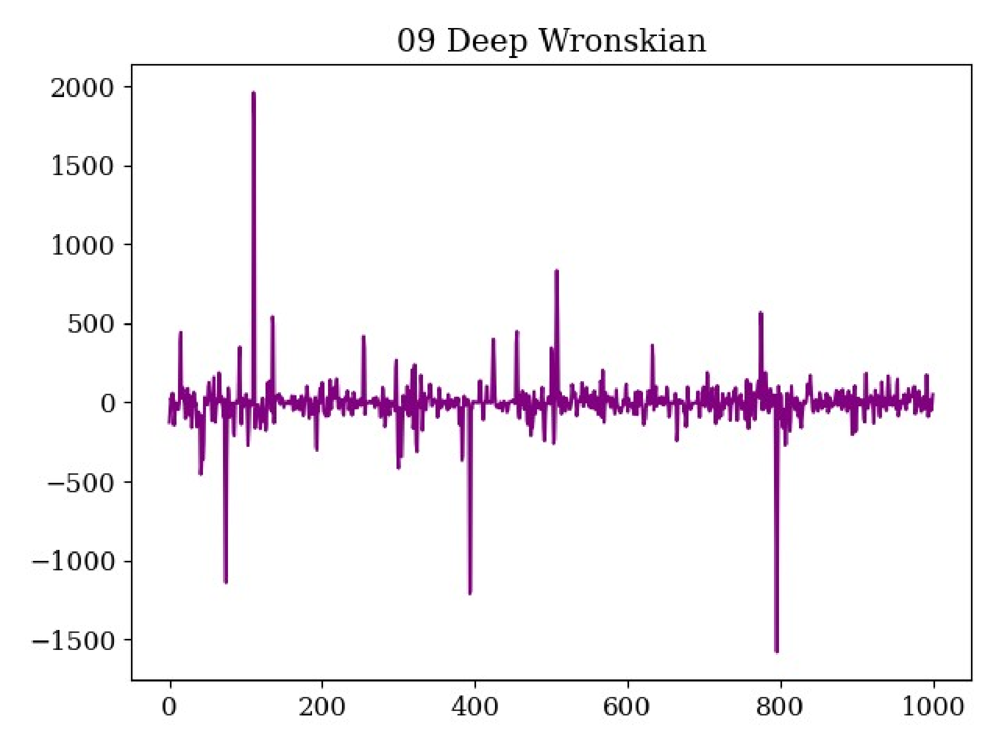
- P′(t) tells how price is changing.
- S′(t) tells how spread is changing.
- If W(t) stays exactly zero, price and spread move like dependent functions.
- If W(t) fluctuates away from zero, price and spread contain different information.

Example: price rises while spread narrows. Then price and spread are clearly describing different market effects, so W(t) is non-zero.

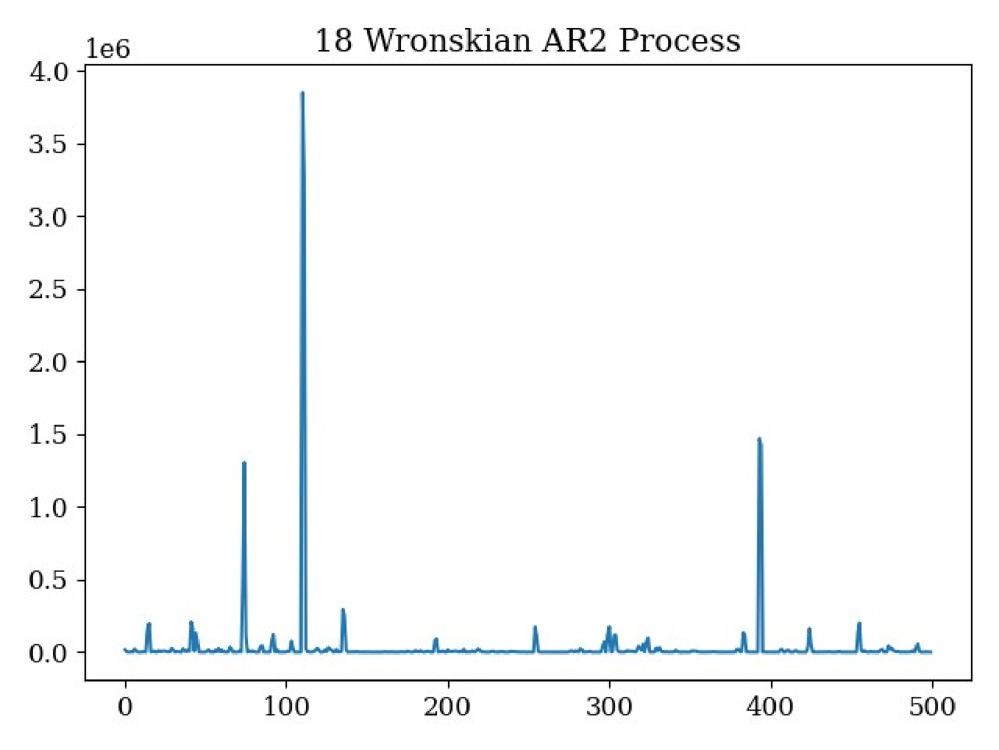
Project result:
|W_avg| = 2.5902
Std. deviation = 211.59

Wronskian graphs: independence plus shock bursts

The signal is mostly around zero but repeatedly moves away from zero.



Deep Wronskian W(t) = P(t)S′(t) − S(t)P′(t)



AR(2) spectral behaviour of the Wronskian process

- Left: W(t) never settles identically at zero, so price and spread are not redundant.
- Large spikes mark moments where the relation between price and spread changes sharply.
- Right: the AR(2)/spectral view shows rare bursts of very high energy.
- Trading use: Wronskian spikes can act as warnings of structural signal stress.

20-dimensional homogeneous linear ODE system

Now the whole 10-bid + 10-ask book is modeled together.

$$\dot{x}(t) = A\ x(t)$$

- x(t) is a 20-dimensional vector of order-book depths.
- The first 10 components represent bid-side levels.
- The next 10 components represent ask-side levels.
- A is a 20 × 20 interaction matrix: each entry says how one level affects another.

Why this matters: Level 1 does not move alone. A shock at one depth level can ripple across near and far liquidity.

$$x(t) = c_1 e\hat{}(\lambda_1 t) v_1 + \ldots + c_{20} e\hat{}(\lambda_{20} t) v_{20}$$

Each eigenvalue λ describes one independent mode of book movement.

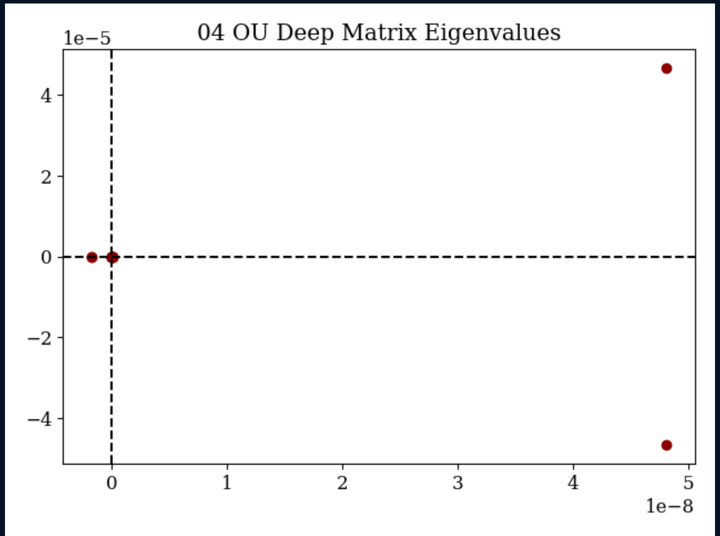
Eigenvalues tell us stability vs breakout

The sign of the real part decides whether a mode decays or grows.

$$\mathrm{Re}(\lambda) < 0 \;\rightarrow \text{disturbance decays} \;\rightarrow \text{stable / mean-reverting}$$

$$\mathrm{Re}(\lambda) > 0 \;\rightarrow \text{disturbance grows} \;\rightarrow \text{unstable / drifting}$$

- The report finds negative eigenvalues near −0.0003: these suppress top-of-book deviations.
- It also finds positive eigenvalues near +0.0032: these allow deep-book pressures to grow.
- Because both signs appear, the system is saddle-like: some directions return to balance while others move away.
- Financial meaning: top levels may look stable, but deeper order blocks can still create momentum.

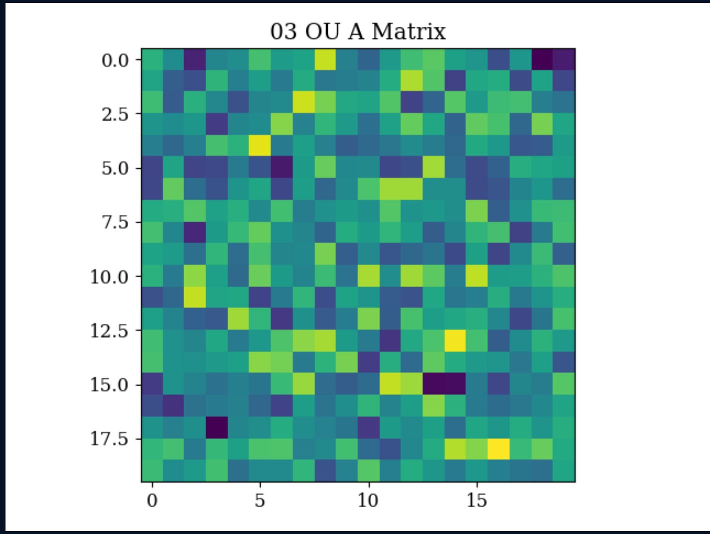


Eigenvalue plot of A

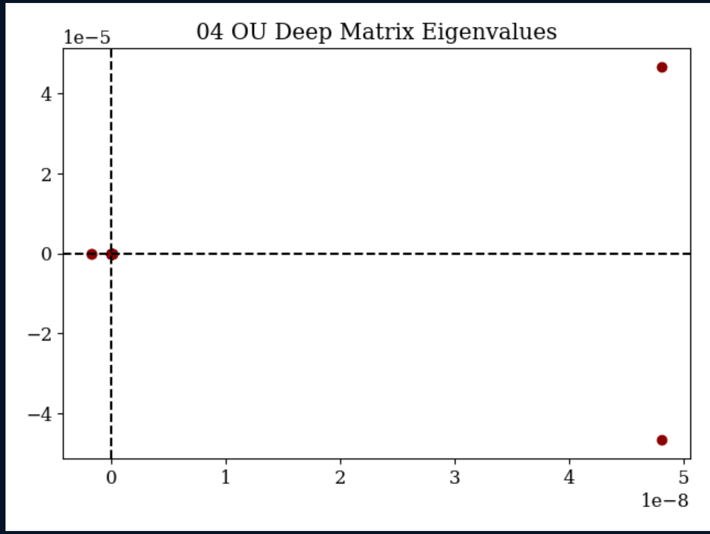
In presentation: “Eigenvalues are the stability DNA of the order book.”

Matrix, eigenvalues, and impulse response

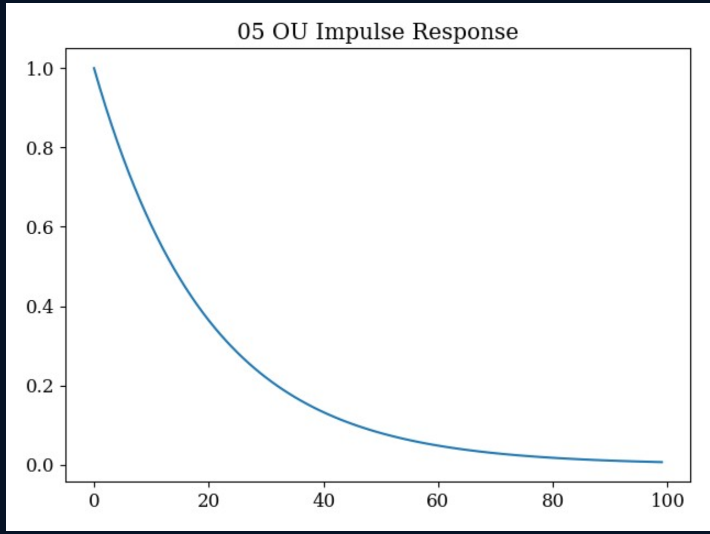
These three graphs show coupling, stability, and recovery speed.



A matrix heat map



Eigenvalues



Impulse response

- Heat map: each square is how one book tier reacts to another tier.
- Eigenvalue plot: positive and negative modes coexist, giving saddle-like behaviour.
- Impulse response: a stable shock decays back toward zero over time.
- Together: local liquidity can be absorbed, but some deep-book modes can still drive trends.

Picard's method: solving an ODE by repeated guessing

Turn the differential equation into an integral equation, then refine repeatedly.

$$y' = f(t,y), \qquad y(t_0)=y_0$$

$$y(t) = y_0 + \int[t_0 \text{ to } t]\ f(s,\, y(s))\ ds$$

- Start with a simple first guess: $\varphi_0(t) = y_0$.
- Plug φ_0 into the integral to get φ_1.
- Then plug φ_1 into the integral to get φ_2.
- Keep repeating until the curves stop changing.

Analogy: it is like repeatedly correcting a forecast until the forecast becomes stable.

In the project, the function f(t,y) comes from empirical market flow, and y(t) represents short-horizon mid-price movement.

Picard recursion in the project

A transparent alternative to black-box short-horizon forecasting.

$$\varphi_0(t) = y_0$$

$$\varphi_1(t) = y_0 + \int f(s, \varphi_0(s))\ ds$$

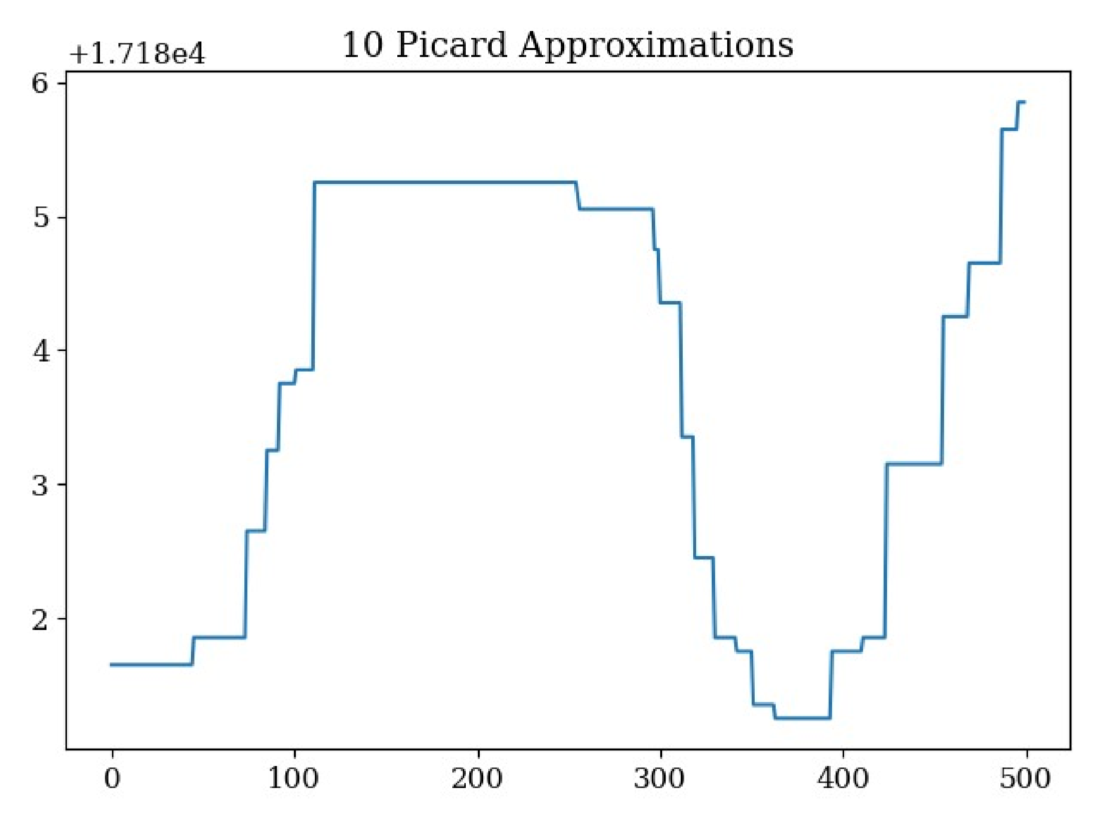
$$\varphi_{n+1}(t) = y_0 + \int f(s, \varphi_n(s))\ ds$$

Project interpretation: classical ODE recursion behaves like an explainable unrolled sequence model.

- Each iteration uses the previous curve to create a better curve.
- If the market is locally smooth, the iterations converge.
- If a regime shock breaks smoothness, convergence fails or error explodes.
- That failure becomes a warning signal for unpredictable conditions.

Picard graph: convergence of the approximation

The project simulates repeated Picard integrations on a volatile window.



10 Picard approximations

• The curve shows successive approximations across a short price window.
• The report says 12 integrations reduce relative error from about 6.5 to 0.074.
• When the error shrinks, the ODE path is locally predictable.
• If the error stops shrinking, the market may have entered a shock regime.

Presentation line: “Picard gives both a forecast and a warning: convergence means stable local dynamics; divergence means the model should not trust the path.”

Cauchy–Euler equations: power-law market memory

Some market shocks fade slowly, not exponentially.

- A simple exponential model says shocks disappear quickly at a fixed decay rate.
- But financial markets often remember shocks for longer than that.
- Cauchy–Euler ODEs naturally produce power-law solutions of the form y(t) = t^r.
- This is useful for spread shocks, volatility shocks, and post-event liquidity gaps.

$$\text{exponential decay:}\ \ \mathrm{Ce\hat{}(-kt)}$$

$$\text{power-law decay:}\ \ \mathrm{C\ t\hat{}(-\mu)}$$

Power-law decay is slower in the tail. That means a shock can keep affecting spreads and volatility much longer than a simple exponential model predicts.

Cauchy–Euler formulation

The coefficients scale with time, so the solution is searched as y = t^r.

$$a_n t^n y^{(n)} + a_{n-1} t^{n-1} y^{(n-1)} + \ldots + a_1 t y' + a_0 y = 0$$

$$\text{Second order case:} \quad a\ t^2 y'' + b\ t\ y' + c\ y = 0$$

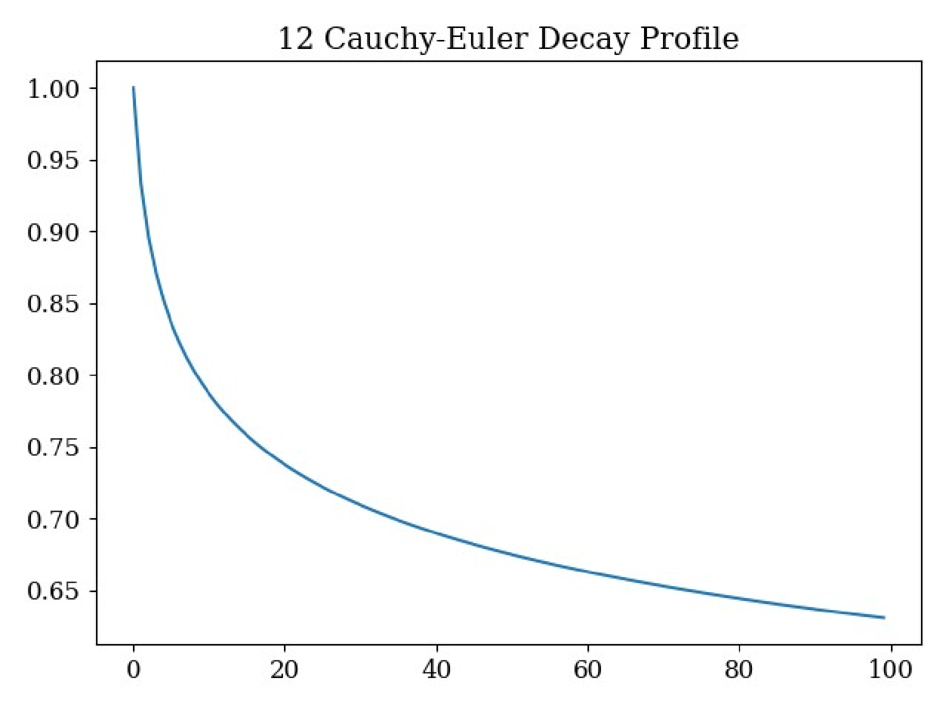
$$\text{Assume } y(t)=t\hat{}r \ \rightarrow \ ar(r-1)+br+c=0$$

- Solving the quadratic gives the exponent r.
- That exponent tells how fast shock impact decays.
- The project reports roots around r ≈ 0.94.
- Interpretation: impact fades sub-linearly and remains longer than simple exponential half-life assumptions.

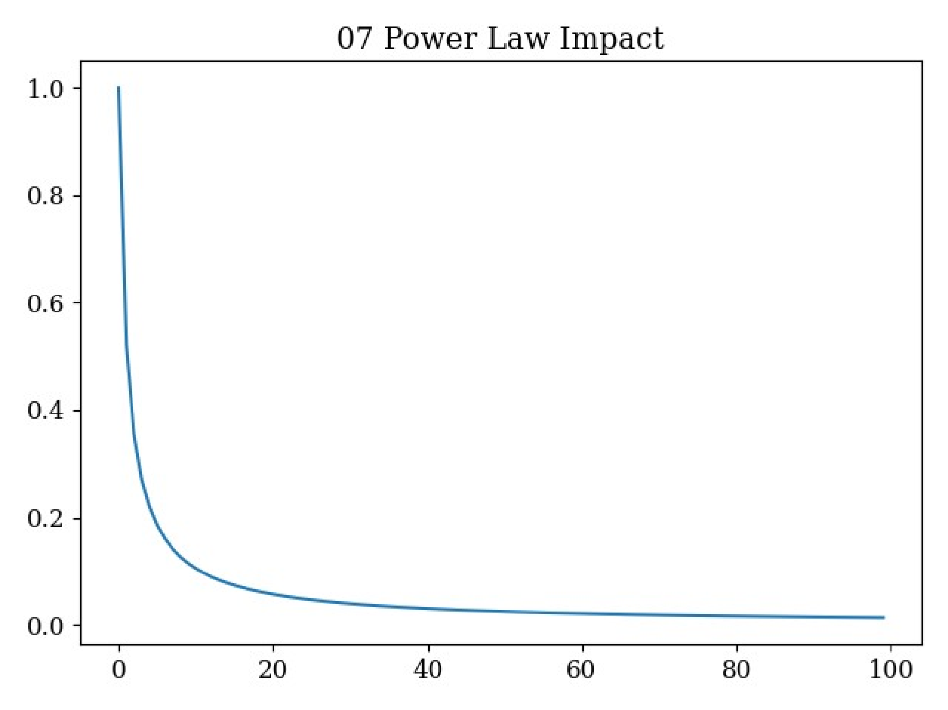
For presentation: “r is the memory exponent of the market shock.”

Cauchy–Euler graphs: slow post-shock decay

Both plots show long-tail behaviour after a market impact.



Spread shock decay profile



Power-law price impact

- Left: the spread shock falls quickly at first, then decays slowly.
- Right: price impact also has a long tail, consistent with power-law memory.
- Trading implication: after a shock, market makers should not tighten spreads too early.
- Risk implication: recent shocks remain relevant for longer than exponential models suggest.

Exact differential equations: finding low-slippage paths

The project treats execution as movement through a market field.

$$\mathrm{M(x,y)\ dx + N(x,y)\ dy = 0}$$

$$\text{exact if}\ \ \partial \mathrm{M}/\partial \mathrm{y} = \partial \mathrm{N}/\partial \mathrm{x}$$

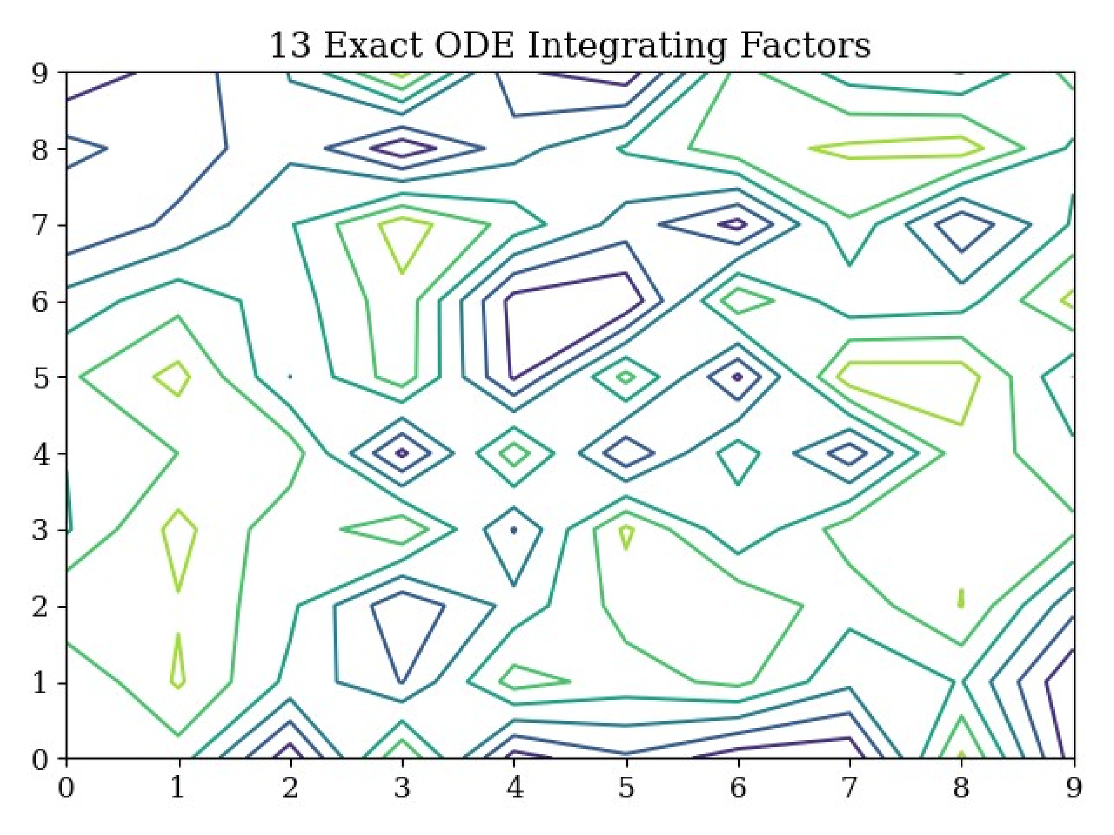
- If a differential form is exact, it has a scalar potential Ψ(x,y).
- Moving along potential contours means path-independent behaviour.
- In market terms, this is interpreted as lower rotational slippage/friction.
- If the field is not exact, an integrating factor μ may transform it into an exact form.

Project variables:
x = executed volume
y = local spread expansion

Claim: better routing follows near-conservative paths, reducing unnecessary slippage.

Exact-field graph: mapping slippage topology

The contour plot visualizes the integrating-factor field.



Exact ODE integrating factors / potential contours

- The contour lines represent transformed market-field structure.
- Initially, the report finds Clairaut/exactness violations around 2.201.
- After integrating-factor correction, RMS rotational variance falls to about 0.173.
- Financial meaning: the execution field becomes closer to conservative, so routing can avoid excess slippage.

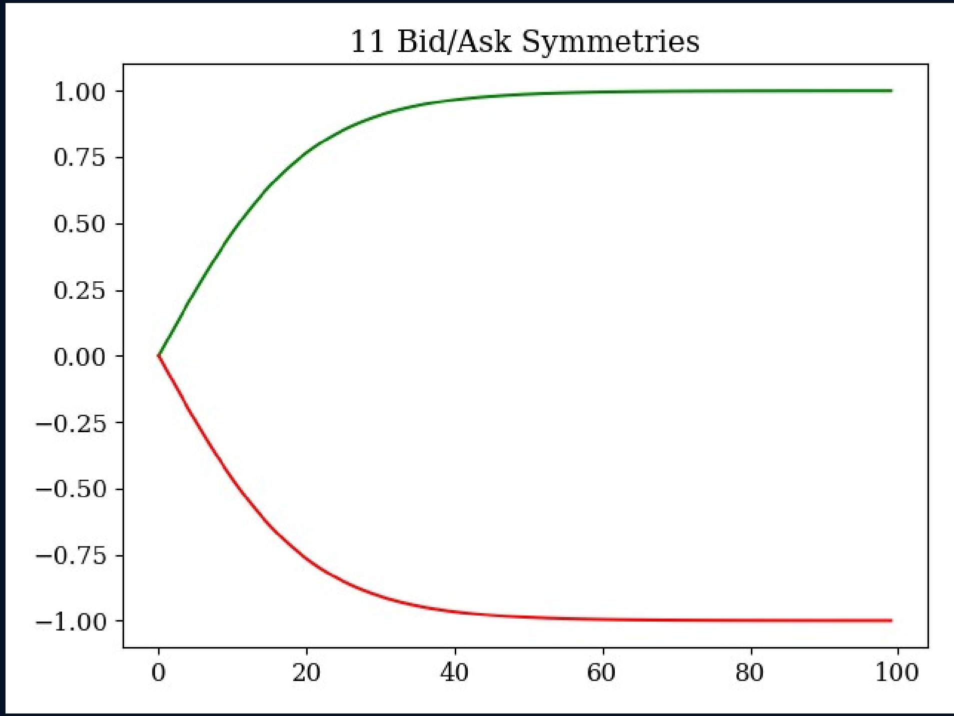
Simple line: “This section turns execution cost into a geometry problem.”

Symmetric operators: bid and ask as mirror systems

The project asks whether buy pressure and sell pressure behave like opposites.

$$\text{symmetry condition:} \quad f(-y) = -f(y)$$

- If the system is symmetric, then flipping the sign of pressure should flip the response.
- Buy-side pressure and sell-side pressure should be mirror images under normal conditions.
- This helps justify using similar strategy logic for long and short sides.
- The project reports strong bid/ask symmetry evidence with very small KS p-values.

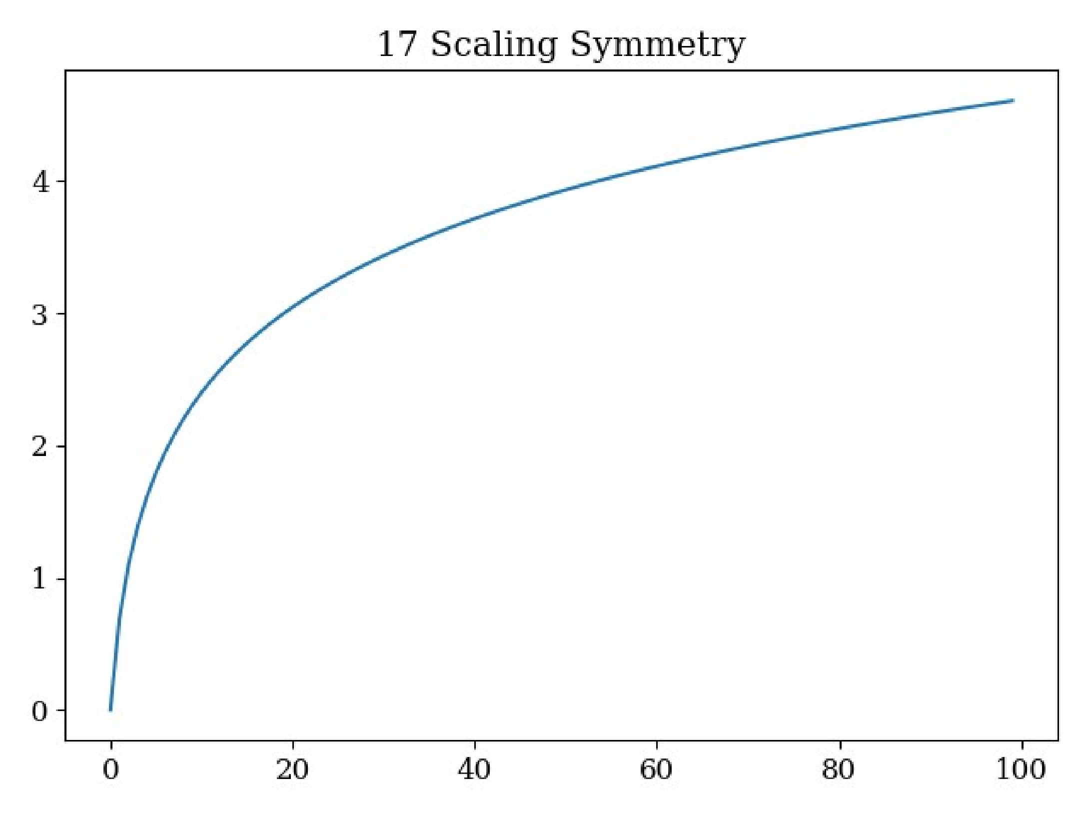


Bid/ask mirror symmetry

The green and red curves behave like mirrored versions of each other.

Scale symmetry: behaviour across market size levels

The project also checks whether structure remains similar after scaling.



Scaling symmetry feature

- Scale symmetry means the same type of structure can appear at different magnitudes.
- In markets, small and large liquidity changes may follow similar shapes after normalization.
- The graph has a smooth logarithmic-looking rise, suggesting scale-invariant behaviour.
- Practical use: models can sometimes generalize from smaller events to larger events after appropriate scaling.

Together, mirror symmetry and scale symmetry support reusable long/short and small/large market logic.

Lipschitz bounds: when does a unique price path exist?

Picard–Lindelöf gives the mathematical condition for predictable local evolution.

$$|f(t,y_1) - f(t,y_2)| \leq L\ |y_1 - y_2|$$

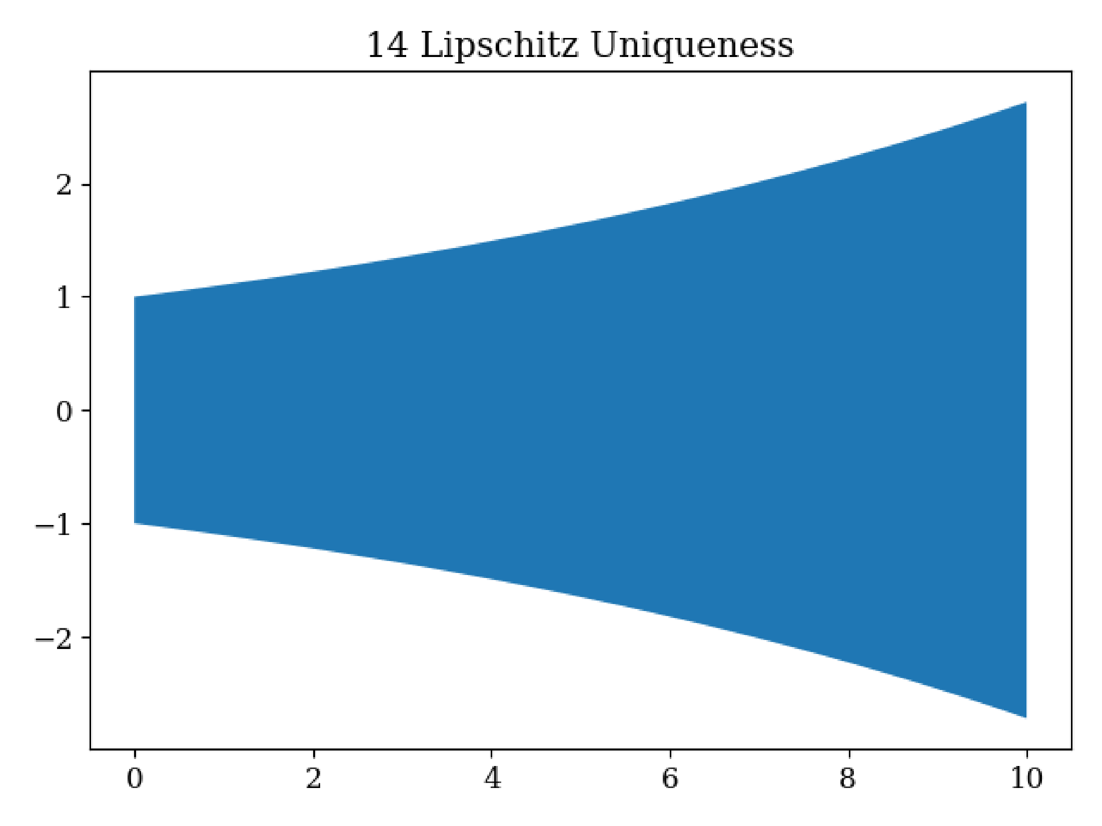
• The function f must be continuous and not too steep in y.
• L is the Lipschitz constant: it controls how sensitive the system is to tiny changes.
• If the bound holds, there is one unique local solution path.
• If the bound breaks, small changes can produce very different price paths.

Financial meaning:
Normal market → unique predictable path.
Flash crash / shock → Lipschitz bound can fail, so prediction becomes fragile.

This connects directly back to Picard iteration: convergence requires this type of smoothness.

Lipschitz graph: a widening uncertainty region

The visual shows how allowed movement can expand when sensitivity increases.



Lipschitz uniqueness bound

- The shaded region widens as time increases.
- Wider bounds mean small differences in initial state can lead to larger differences later.
- In calm markets, L is controlled and the solution path is unique enough to forecast.
- In shock markets, L can spike; then deterministic forecasting becomes unreliable.

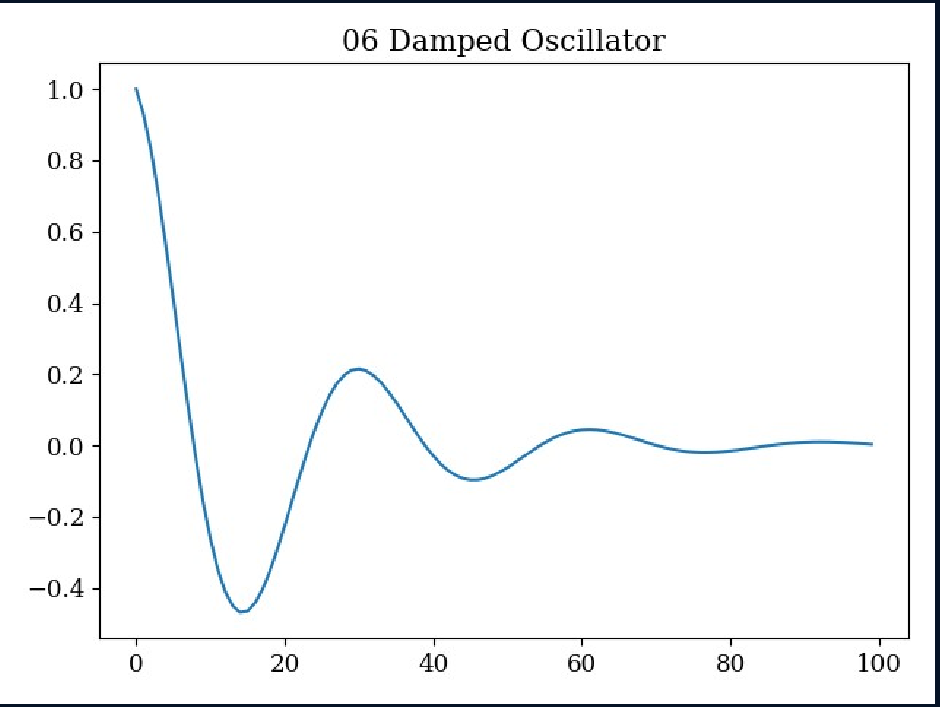
Trading response: when the uniqueness bound becomes unstable, reduce inventory, widen quotes, or stop relying on the short-term path.

Spring-mass analogy: price as a damped oscillator

A second-order ODE can describe momentum, damping, and reversion.

$$m\ x'' + c\ x' + k\ x = F\_ext(t)$$

- m: inertia — large market capitalization or heavy depth slows reaction.
- c: damping — liquidity absorbs velocity and reduces oscillation.
- k: elastic reversion — force pulling price back toward equilibrium.
- F_ext(t): external shock or order-flow pressure.

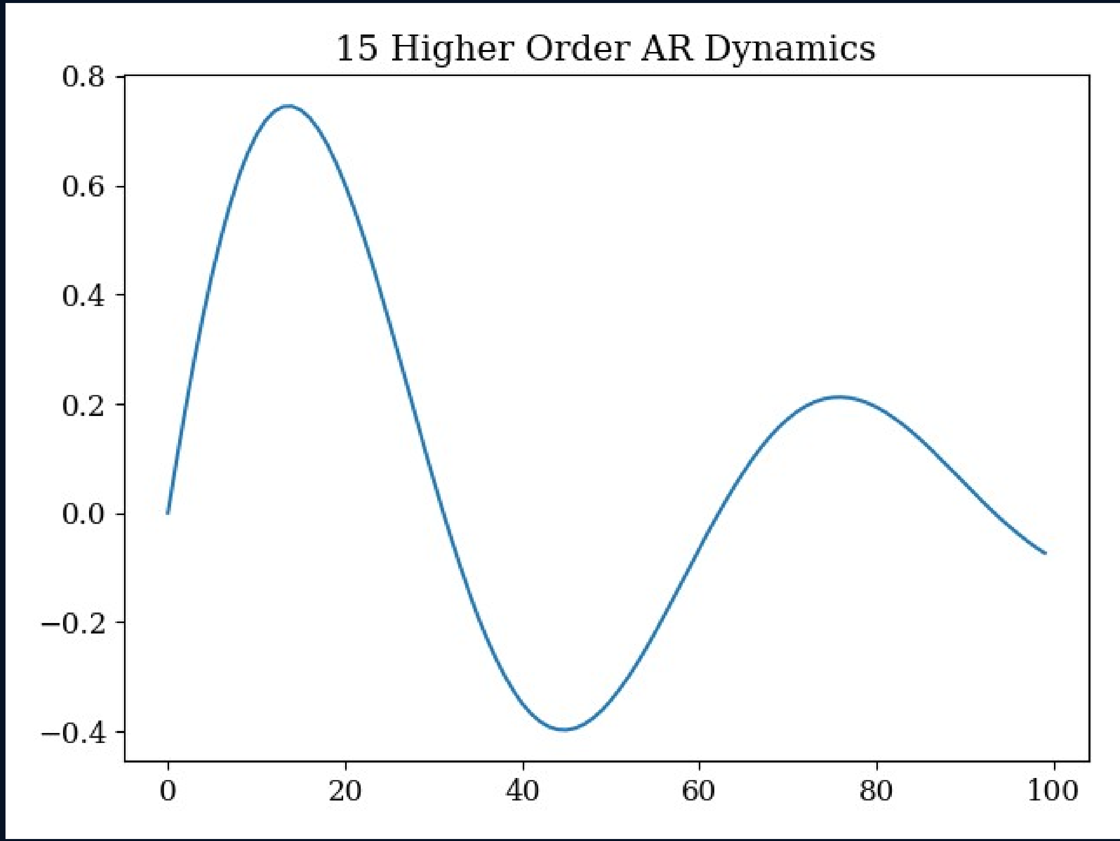


Damped oscillator response

Interpretation: after a shock, price can overshoot, reverse, and gradually settle — just like a damped mechanical system.

Higher-order dynamics: oscillation beyond simple damping

The project also visualizes higher-order autoregressive behaviour.



Higher-order AR dynamics

• The curve shows repeated waves rather than one clean decay.
• This suggests market response can contain delayed feedback loops.
• In order books, algorithms react to algorithms, creating oscillatory microstructure patterns.
• Risk meaning: even after price appears to revert, secondary waves can still appear.

Presentation line: “Momentum does not always stop at equilibrium; it can overshoot, correct, and oscillate.”

How the report turns formulas into graphs

This section explains the implementation pipeline used on the order-book data.

1	**Raw snapshots**	Bid/ask prices and volumes are read at every timestamp.
2	**Variables**	Compute mid-price, spread, OBI, depth vectors, and liquidity volume.
3	**Derivatives**	Use finite differences: value change divided by Δt.
4	**Fit / calculate**	Apply the ODE formula: fit parameters, compute Wronskians, eigenvalues, or iterations.
5	**Plot**	Graph the resulting phase points, fitted curves, spectra, impulse responses, or residual errors.

$$\text{derivative} \approx (\text{next value} - \text{current value}) \,/\, \Delta t$$

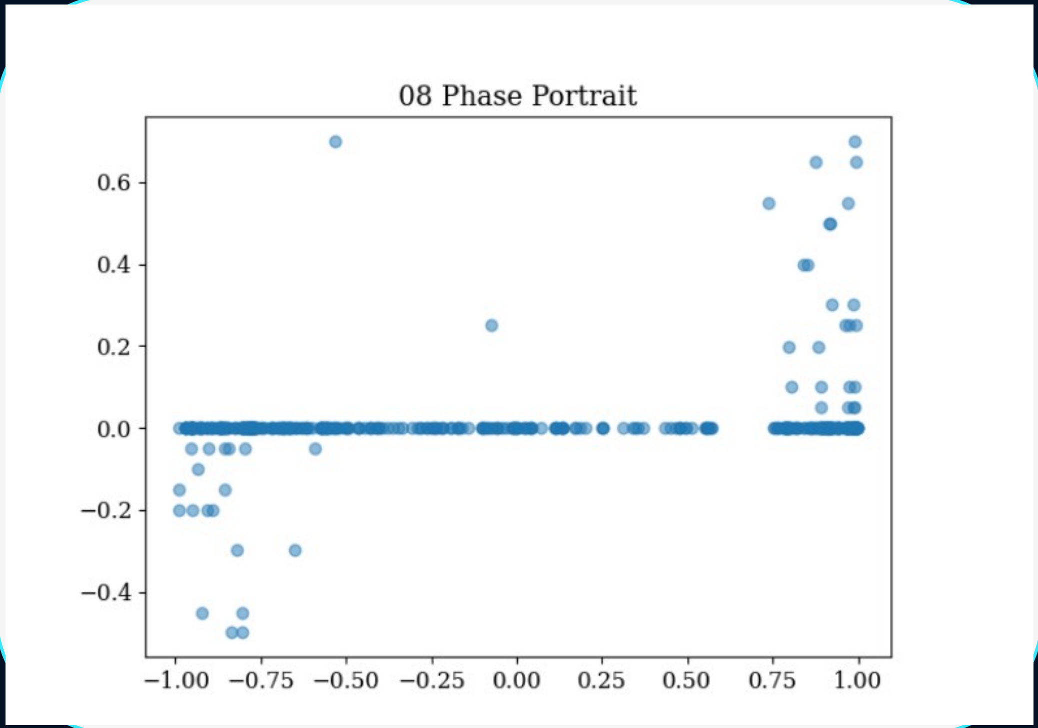
Phase portrait: how the graph was created

The graph is not drawn directly from theory; it is drawn from computed OBI and velocity values.

$$\mathrm{OBI}_t = (\mathrm{BidVol}_t - \mathrm{AskVol}_t)\ /\ (\mathrm{BidVol}_t + \mathrm{AskVol}_t)$$

$$\mathrm{Velocity}_t \approx (\mathrm{MidPrice}_{t+1} - \mathrm{MidPrice}_t)\ /\ \Delta t$$

- For every timestamp, compute one point: (OBI_t, $\mathrm{Velocity}_t$).
- Plot all points together to see calm regions and breakout regions.
- The dense band around velocity = 0 means most ticks have little price movement.
- Extreme OBI values near ±1 create the visible velocity spikes.



Generated from OBI and mid-price velocity

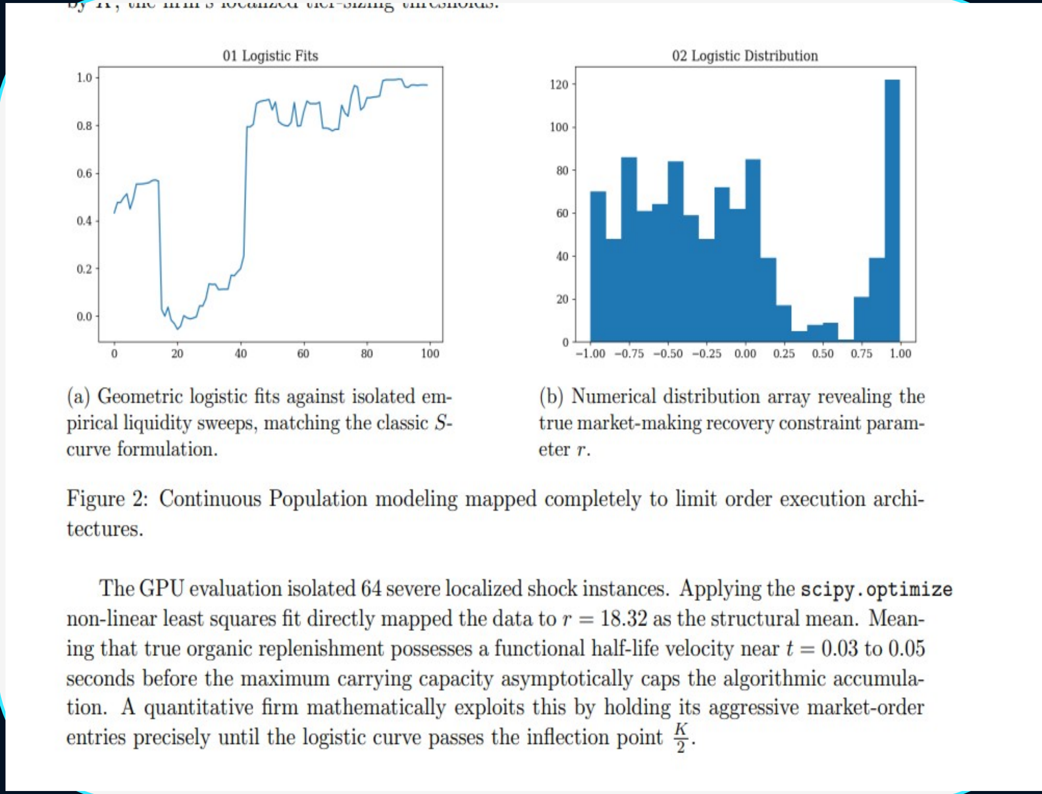
Formula → Data use: each row becomes one coordinate in phase space.

Logistic recovery: how the fit graph was created

The formula was fitted to liquidity after shocks.

$$V(t) = K\ /\ \{1 + [(K - V_0)/V_0]\ e\hat{\ }(-rt)\}$$

- Find shock windows where liquidity suddenly drops.
- For each window, measure total available book volume V(t) after the drop.
- Fit the logistic curve to the recovery using non-linear least squares.
- The fitted parameter r becomes the liquidity recovery speed.
- The report’s fitted mean is r = 18.32, meaning very fast recovery.



(a) Geometric logistic fits against isolated empirical liquidity sweeps, matching the classic *S*-curve formulation.

(b) Numerical distribution array revealing the true market-making recovery constraint parameter r.

Figure 2: Continuous Population modeling mapped completely to limit order execution architectures.

The GPU evaluation isolated 64 severe localized shock instances. Applying the `scipy.optimize` non-linear least squares fit directly mapped the data to $r = 18.32$ as the structural mean. Meaning that true organic replenishment possesses a functional half-life velocity near $t = 0.03$ to 0.05 seconds before the maximum carrying capacity asymptotically caps the algorithmic accumulation. A quantitative firm mathematically exploits this by holding its aggressive market-order entries precisely until the logistic curve passes the inflection point $\frac{K}{2}$.

Shock → refill → saturation

$$\text{fastest refill happens near } V = K/2$$

Wronskian: how independence was computed

Price and spread were tested as two separate signals.

$$W_t = P_t\, S'_t - S_t\, P'_t$$

- Compute mid-price P_t and spread S_t for every tick.
- Approximate P'_t and S'_t using central differences.
- Calculate W_t at every timestamp.
- Plot W_t over time: if it never collapses to zero, the features are not redundant.
- This produced the “Deep Wronskian” plot.

4 3. Wronskian Determinants and Linear Independence

4.1 Rigorous Mathematical Formulation

When processing multi-variate predictor variables (features) for systematic trading rules, a critical mathematical breakdown occurs if the signals share perfect collinearity. To conclusively prove functional non-redundancy along continuous manifolds, we deploy the Wronskian test.

Definition 4.1. *Let $f_1, f_2, \ldots, f_n$ be $(n-1)$ times differentiable functions operating dynamically on a continuous temporal interval I. Their geometric Wronskian $W(f_1, \ldots, f_n)(t)$ is firmly established as:*

$$W(t) = \begin{vmatrix} f_1(t) & f_2(t) & \cdots & f_n(t) \\ f_1'(t) & f_2'(t) & \cdots & f_n'(t) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(t) & f_2^{(n-1)}(t) & \cdots & f_n^{(n-1)}(t) \end{vmatrix} \tag{11}$$

Theorem 4.1 (Non-vanishing Theorem). *If $W(t_0) \neq 0$ efficiently for at least one critical $t_0 \in I$, the functional solutions $\{f_1, \ldots, f_n\}$ are strictly, perfectly linearly independent across I.*

4.2 Feature Defensibility in Crypto Pair Models

In trading, the f_1 vector might be the continuous Price $P(t)$, whilst f_2 is the ongoing bid-ask Spread dimension $S(t)$. We utilized central-difference numerical derivation heavily across PyTorch CUDA blocks to establish the base calculus mappings $P'(t)$ and $S'(t)$, mapping the subsequent determinant $W = P(t)S'(t) - S(t)P'(t)$ continuously spanning 3,730,870 micro units.

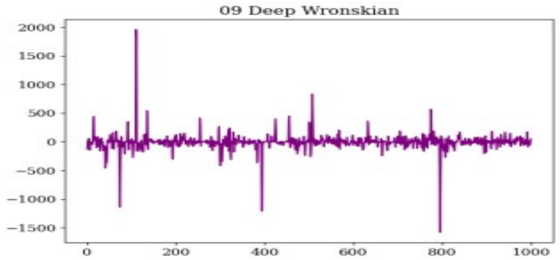


Figure 3: Continuous fluctuations showcasing the dense localized Wronskian determinant oscillating. W never stabilizes mechanically at identically zero.

The mean amplitude of the Wronskian resides at $|W_{avg}| = 2.5902$, maintaining massive spatial separation from 0 alongside a high standard deviation of 211.59. Consequently, P and S absolutely represent uncorrelated signal vectors under an ODE paradigm.

Wronskian fluctuations over time

Interpretation: price and spread carry different information.

Wronskian AR(2) spectrum: how the spike graph was produced

After computing W_t, the project studies whether its shock bursts have structure.

$$W_t \approx a_1 W_{t-1} + a_2 W_{t-2} + \varepsilon_t$$

- Use the Wronskian time series from the previous slide.
- Fit an AR(2) model using two previous W values.
- Convert the fitted process into a spectral/power view.
- Tall spikes represent rare frequencies/lags where the Wronskian energy becomes very large.
- This supports the idea of mostly stable behaviour with occasional microstructure shocks.

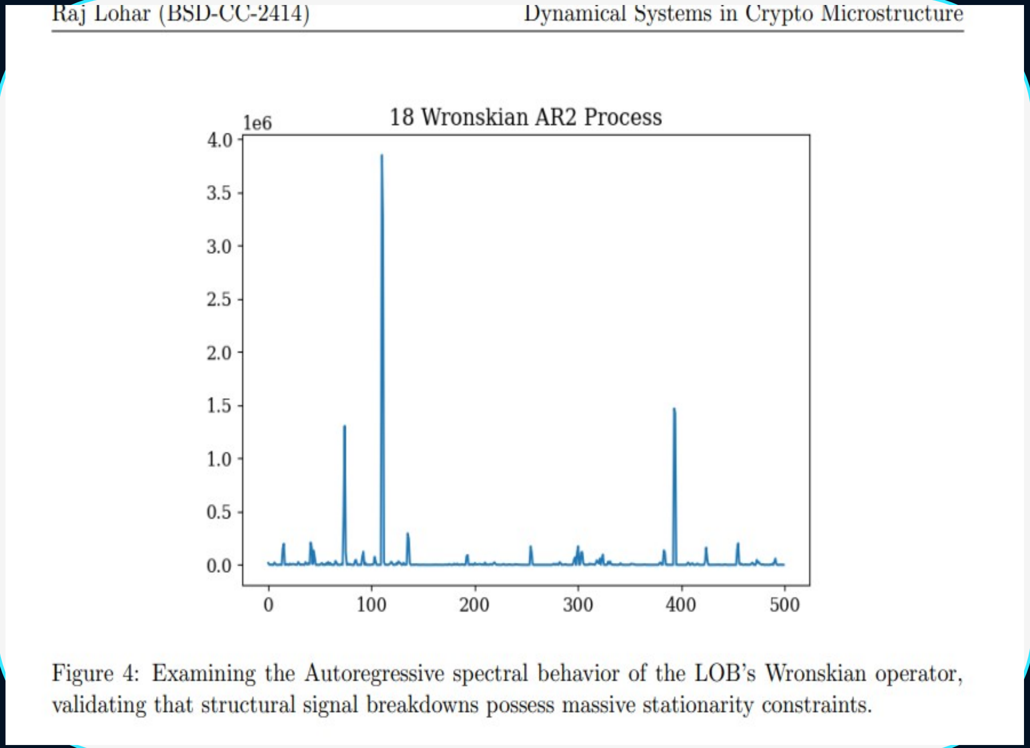

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Figure 4: Examining the Autoregressive spectral behavior of the LOB's Wronskian operator, validating that structural signal breakdowns possess massive stationarity constraints.

Spectral spikes in the Wronskian operator

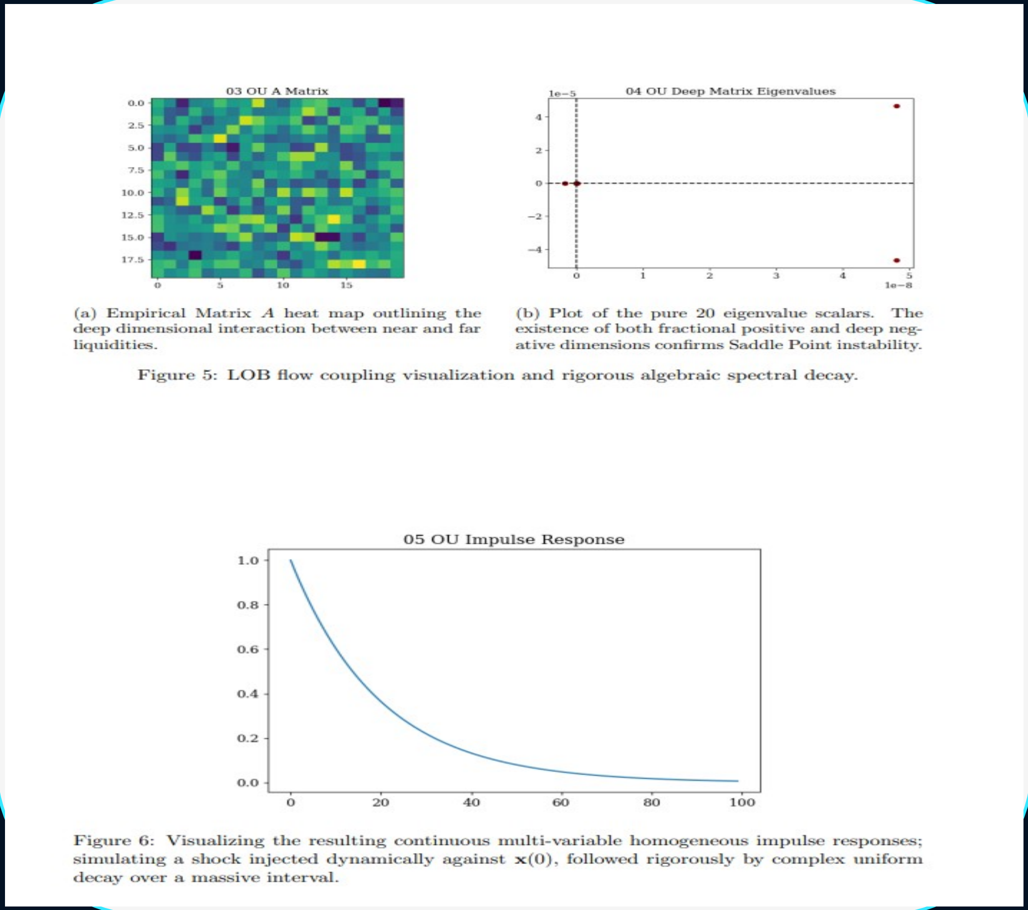
20-level linear ODE: how matrix and eigenvalue graphs were created

Each timestamp becomes a 20-dimensional depth vector.

$$x(t) = [\mathrm{BidVol}_1 \ldots \mathrm{BidVol}_{10},\ \mathrm{AskVol}_1 \ldots \mathrm{AskVol}_{10}]^T$$

$$\dot{x}(t) \approx [x(t+\Delta t) - x(t)] \,/\, \Delta t \approx A\, x(t)$$

- Build $x(t)$ from 10 bid volumes and 10 ask volumes.
- Calculate $\dot{x}(t)$, the change in the depth vector.
- Fit matrix A so that $A\,x(t)$ best explains $\dot{x}(t)$.
- Plot A as a heat map to show coupling between levels.
- Compute eigenvalues of A to separate stable and unstable modes.



A matrix, eigenvalues, and impulse response

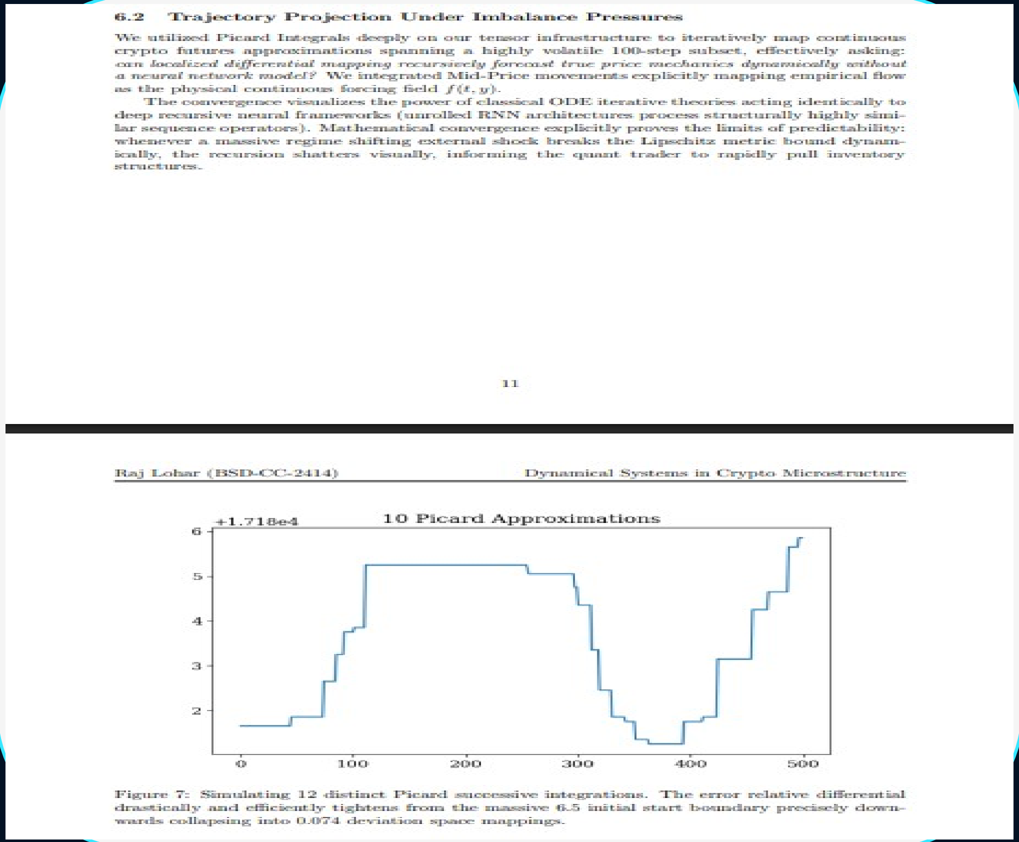
Negative eigenvalue → decay; positive eigenvalue → drift/breakout.

Picard approximation: how the convergence graph was created

The method repeatedly improves a short-horizon price path.

$$\varphi_{n+1}(t) = y_0 + \int f(s, \varphi_n(s))\, ds$$

- Choose a volatile 100-step price window.
- Start with a simple initial path: $\varphi_0(t)=y_0$.
- Use the empirical order-flow force $f(t,y)$ to integrate a better path.
- Repeat the integral update 12 times.
- Plot approximation/residual values to show convergence.



6.2 Trajectory Projection Under Imbalance Pressures

We utilized Picard Integrals deeply on our tensor infrastructure to iteratively map continuous crypto futures approximations spanning a highly volatile 100-step subset, effectively asking: *can localized differential mapping recursively forecast true price mechanics dynamically without a neural network model?* We integrated Mid-Price movements explicitly mapping empirical flow as the physical continuous forcing field $f(t, y)$.

The convergence visualizes the power of classical ODE iterative theories acting identically to deep recursive neural frameworks (unrolled RNN architectures process structurally highly similar sequence operators). Mathematical convergence explicitly proves the limits of predictability: whenever a massive regime shifting external shock breaks the Lipschitz metric bound dynamically, the recursion shatters visually, informing the quant trader to rapidly pull inventory structures.

11

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Figure 7: Simulating 12 distinct Picard successive integrations. The error relative differential drastically and efficiently tightens from the massive 6.5 initial start boundary precisely downwards collapsing into 0.074 deviation space mappings.

Successive Picard approximations

Data result: residual error shrinks from about 6.5 to 0.074.

Cauchy–Euler shock memory: how power-law graphs were created

The project tests whether shock effects decay slowly instead of exponentially.

$$\text{assume shock response } y(t) \approx C\, t\hat{\ }(-r)$$

- Identify a post-shock window: spread expansion or volatility impact after a large event.
- Normalize the shock response so the starting value is comparable across events.
- Fit a power-law curve instead of a simple exponential curve.
- Estimate the exponent r; the report finds r ≈ 0.94.
- Plot the fitted decay profile and impact curve.

Meaning: market memory fades slowly after shocks.

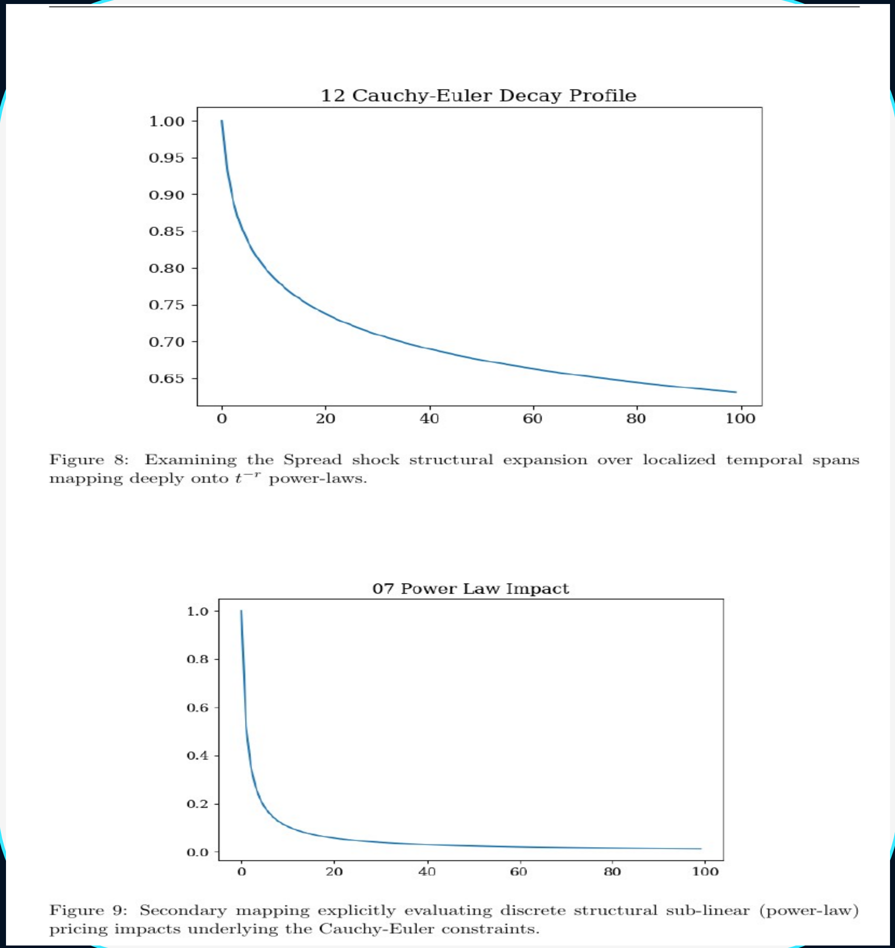


Figure 8: Examining the Spread shock structural expansion over localized temporal spans mapping deeply onto t^{-r} power-laws.

Figure 9: Secondary mapping explicitly evaluating discrete structural sub-linear (power-law) pricing impacts underlying the Cauchy-Euler constraints.

Power-law decay and impact profiles

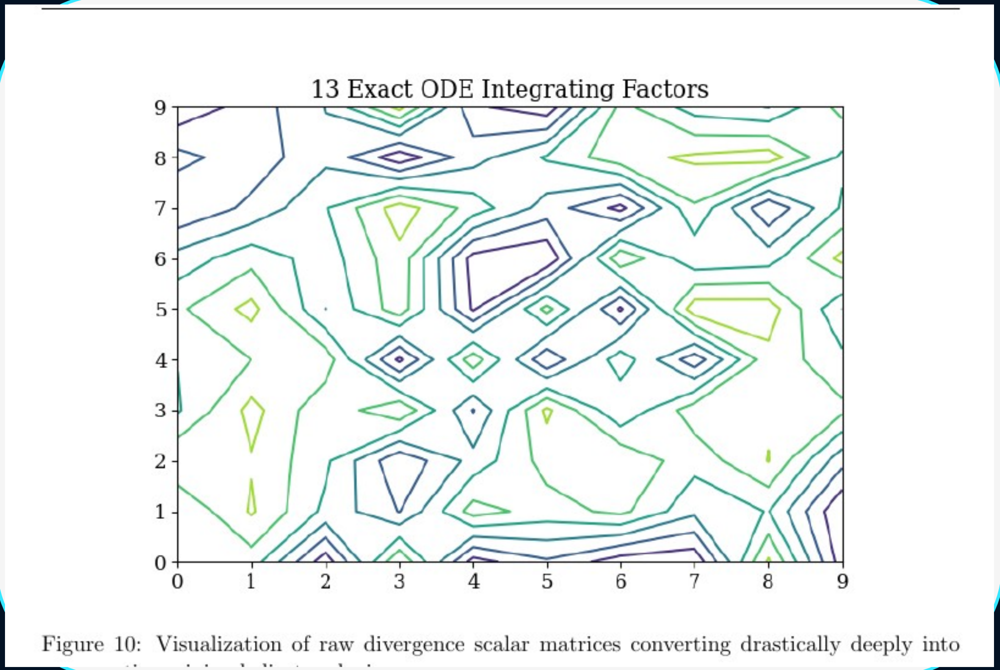
Exact differential field: how the slippage graph was created

Execution volume and spread expansion are treated as a 2D field.

$$M(x,y)\ dx + N(x,y)\ dy = 0$$

$$\text{exact if } \ \partial M/\partial y = \partial N/\partial x$$

- Let x = executed volume and y = local spread expansion.
- Estimate field components M and N from observed execution/spread changes.
- Measure how much the field violates exactness using partial-derivative mismatch.
- Apply an integrating-factor-style correction to reduce rotational friction.
- Plot contour/topology after this transformation.



Exact-field contour view

Data result: divergence/friction scale drops from about 2.201 to 0.173 RMS.

Bid–ask symmetry: how mirror-response graphs were created

The project compares buy pressure with the negative mirror of sell pressure.

$$f(-y) = -f(y)$$

- Separate order-flow responses into bid-side and ask-side pressure series.
- Transform one side by sign inversion to create its mirror version.
- Overlay buy response and mirrored sell response.
- Run statistical comparison to check whether their shapes match.
- The plotted curves show the intended mirror structure.

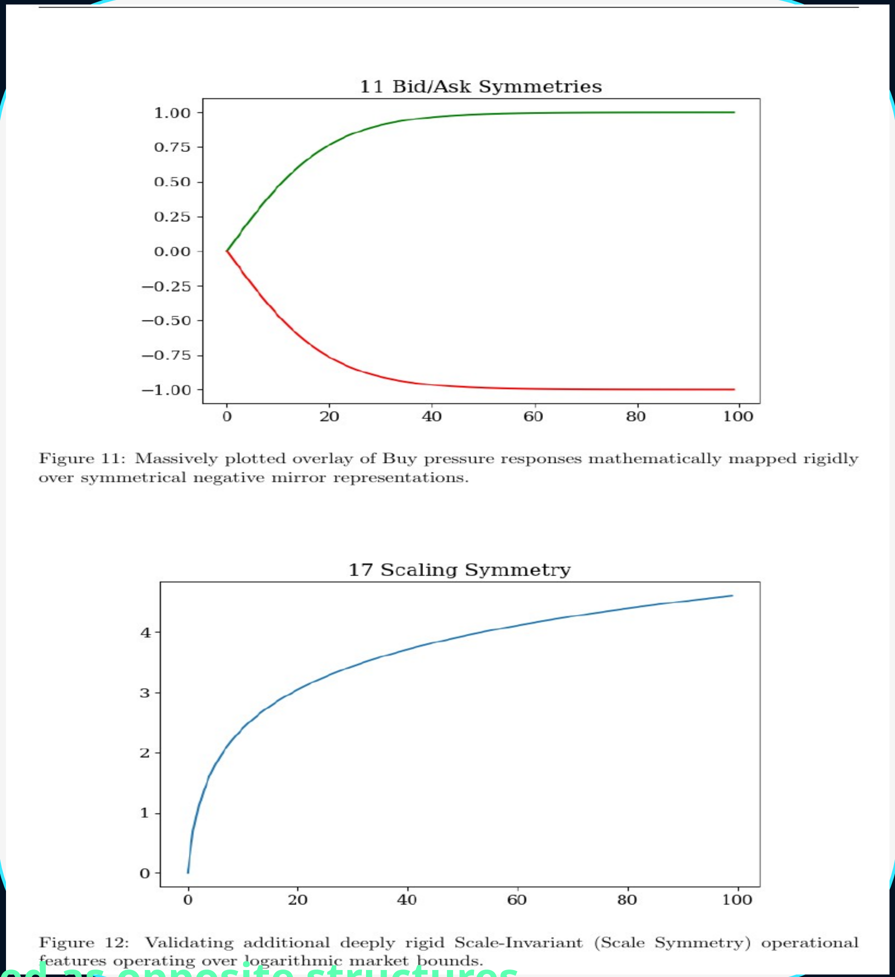


Figure 11: Massively plotted overlay of Buy pressure responses mathematically mapped rigidly over symmetrical negative mirror representations.

Figure 12: Validating additional deeply rigid Scale-Invariant (Scale Symmetry) operational features operating over logarithmic market bounds.

Meaning: long-side and short-side responses can be modeled as opposite structures.

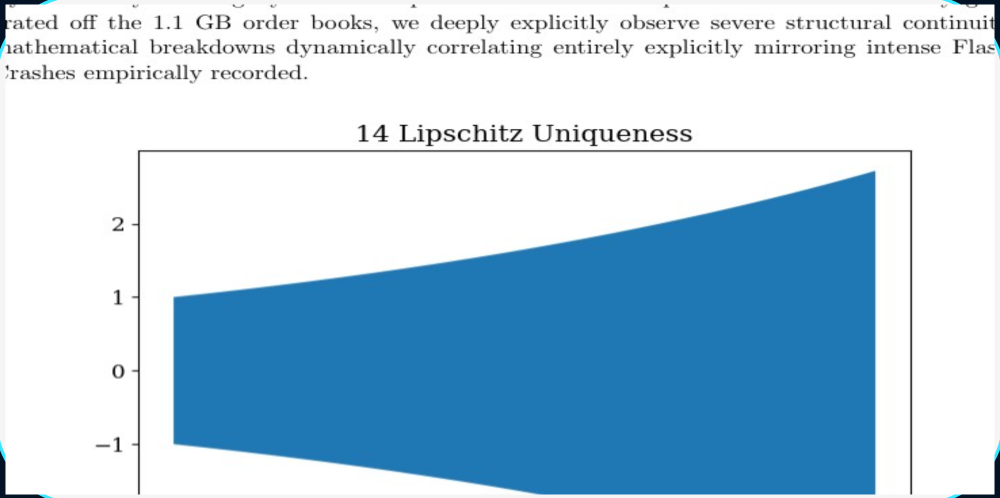
Bid/ask mirror and scale symmetry graphs

Lipschitz and oscillator graphs: how they were produced

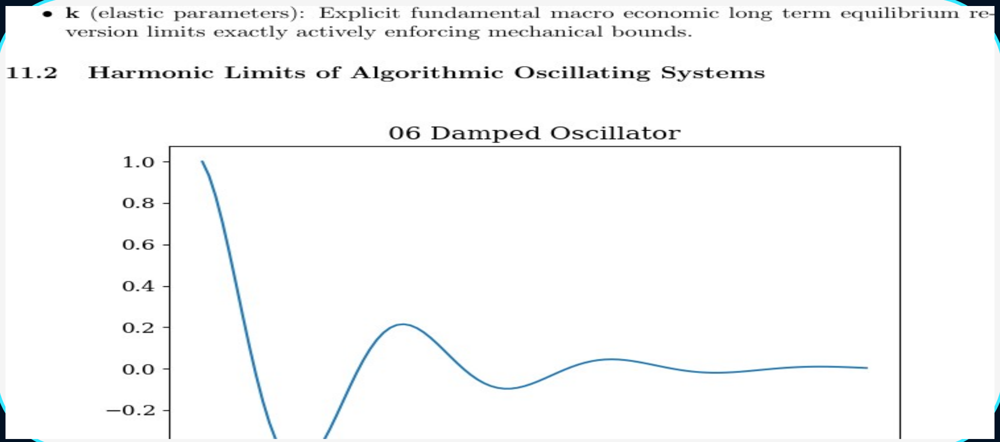
These graphs come from sensitivity tracking and damped-shock response fitting.

$$L_t \approx |f(t,y_1) - f(t,y_2)| \ / \ |y_1 - y_2|$$

• For Lipschitz: compare how strongly nearby market states produce different price forces.
• A widening band means uncertainty is growing; predictability is weakening.
• For oscillator: fit price deviation after a shock to m x″ + c x′ + kx = Fext(t).
• The decaying wave shows overshoot followed by liquidity damping.



Lipschitz uncertainty band



Damped oscillator response

Use: detect when local uniqueness breaks and when shocks are being absorbed.

Complete formula-to-data mapping

Each graph came from turning raw LOB columns into a mathematical object.

Topic	Data input	Graph output
Phase portrait	OBI, MidPrice velocity	scatter of (OBI_t, $Velocity_t$)
Logistic recovery	liquidity after shock	fit V(t) and estimate r
Wronskian	price and spread derivatives	plot $W_t = P_t S'_t - S_t P'_t$
20-level ODE	10 bid + 10 ask volumes	fit A, plot heat map/eigenvalues
Picard	short price window + force field	iterate φ_n and plot convergence
Cauchy–Euler	post-shock spread/volatility	fit power-law t^(−r)
Exact field	executed volume + spread expansion	plot corrected low-friction contour
Symmetry / Lipschitz / Oscillator	bid/ask pressure, sensitivity, shocks	mirror plots, uncertainty bands, damped waves

Main takeaway: every graph is a data-transformed version of an ODE concept, not a separate unrelated plot.

Key empirical findings from the project

These are the numbers to remember from the analysis.

-0.0003 to $+0.0032$	$r = 18.32$	0.074	$r \approx 0.94$	0.173	2.5902
stability spectrum	liquidity recovery	Picard residual	memory exponent	exact-field RMS	Wronskian avg

- Saddle-like behaviour: the book has stable and unstable directions at the same time.
- Liquidity replenishment is very fast after shocks, but capacity-limited.
- Price and spread are not redundant signals under the Wronskian test.
- Shock memory follows a slower power-law pattern, not just exponential decay.
- Exact-field transformations suggest a way to think about lower-slippage routing.

Trading implications in simple language

The math is translated into execution and risk ideas.

- Heal-rate execution: after a liquidity sweep, wait about 40–80 ms spacing so the book can refill.
- Lead-lag imbalance: strong OBI can appear before mid-price movement, so imbalance becomes a short-horizon signal.
- Deep-tier momentum: do not rely only on level-1 noise; deep-book pressure can drive breakout moves.
- Power-law tail risk: after a shock, keep spreads wider longer because memory fades slowly.
- Smart order routing: use conservative/exact-field paths to reduce friction and slippage.

Important: these are research interpretations, not guaranteed trading profits. The model explains mechanisms, but real markets also include stochastic shocks, fees, latency, and execution constraints.

How the ODE tools map to an algorithmic architecture

Each mathematical module can become a signal or risk-control component.

- Phase portrait/eigenvalue module: detect whether the book is in stable reversion or unstable drift.
- Logistic module: estimate post-shock liquidity recovery and decide when to resume execution.
- Wronskian module: monitor whether price and spread remain independent or collapse into signal stress.
- Cauchy–Euler module: estimate how long shock memory should affect quoting and risk limits.
- Lipschitz/Picard module: decide whether short-term trajectory forecasts are currently trustworthy.
- Exact-field module: choose routing paths that minimize avoidable slippage.

Limitations and careful wording

This slide helps you answer questions honestly during presentation.

- The original data is discrete; ODE modeling is a continuous approximation justified by very high tick density.
- Some parameters such as α, β, γ, δ are symbolic in the report; the combined fitted Jacobian/eigenvalues are emphasized more than individual parameter estimates.
- Graphs often use normalized or transformed axes, so they should be explained by behaviour, not only raw units.
- Deterministic ODE tools reveal structure, but they do not remove randomness, news shocks, latency, or execution costs.
- The project should be presented as a mathematical framework for understanding microstructure, not as a finished live-trading system.

Best answer if questioned: “The project shows how ODE methods can organize and explain high-frequency LOB behaviour; further validation would be needed before production trading.”

Conclusion: what the project achieved

A continuous dynamical-systems lens for crypto order books.

- Converted high-frequency crypto LOB data into ODE-style state variables and derivatives.
- Used phase portraits and eigenvalues to study stability and saddle-like drift.
- Used logistic growth to explain bounded liquidity recovery after shocks.
- Used Wronskian determinants to validate signal independence between price and spread.
- Used Picard, Cauchy–Euler, exact equations, symmetry, Lipschitz bounds, and harmonic models to explain prediction, memory, slippage, symmetry, uniqueness, and oscillation.

Final message: the order book is not only a table of orders — at high frequency it behaves like a dynamic physical system with stability, memory, friction, and shock response.

Thank You

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Questions?

This presentation explains the project in simple language while preserving the mathematical structure and original graphs.

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